Fiscal Year:	FY 2026	**Task Last Updated:**	FY 02/03/2026
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Project Title:	Microgravity Biofluid Handling Utilizing RSD Data and Supporting Ground Experiments		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
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Zip Code:	12180-3522	**Congressional District:**	20
Comments:			
Project Type:	Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2023 NNH23ZDA001N-PSI E.8 Physical Sciences Informatics
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No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC25K7630		
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Task Description:	This proposal entails the analysis of in situ fluid behavior to develop first-principle models describing fundamental fluid physics for the control of liquid systems in microgravity. Building upon the late Paul Steen's vision for containerless, and thus frictionless, handling of liquids in microgravity ("Bucket brigade" co-PI Hirsa), this investigation's purpose is to realize the possibility of aqueous systems in microgravity and reduced gravity (e.g., lunar). Biofluids ranging from low concentration native protein solutions to fully fibrillized amyloid gels will be considered for surface-tension containment from sub-millimeter to many centimeters across. One of the greatest challenges for surface tension containment is constraining large liquid packets with minimal solid-surface contact (e.g., thin contact rings), and the often violent dynamics associated with drop breakup (interface creation) and coalescence (interface destruction). Aside from biological, civil, and industrial applications, the anticipated science advancements will be foundational for soft matter and materials science, including extraction and processing of materials (large-scale 3D printing). This project		

	will utilize data from the Ring-Sheared Drop - Amyloid Fibril Formation (RSD-AFF/AFF-R) research on International Space Station (ISS) campaigns and supporting ground-based experiments to produce methodology for analysis of fluid behavior in microgravity and reduced gravity, and to identify primary features of biofluid physics relevant to fluid control in space exploration. The containerless nature of the RSD allows for reactions and material properties to be analyzed without the influence of solid walls on internal dynamics.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Aside from the International Space Station (ISS) and Commercial LEO Destinations (CLDs), there is a need to develop capabilities for sustained human presence at lunar and Martian gravity. [Ed. Note: LEO is an acronym for low Earth orbit.] Understanding and predicting dynamics of biofluids is important to astronauts' health, built environment, production of food, medicines, as well as in-situ resource utilization. If successful, this project extends the late Paul Steen's vision for containerless, and essentially frictionless, handling of liquids at low gravity ("Bucket Brigade" co-PI Hirsa) beyond water, to biofluids.
Task Progress:	Experimental measurement of the RSD's surface profile from drop images at each configuration is essential for performing particle tracking velocimetry (PTV) post-processing, for acquisition of 3D velocity data and for making comparison to computed surfaces. Preliminary results demonstrate that such surface profile determination can be accomplished via the open source state-of-the-art Segment Anything Model 2 (SAM2) and OpenCV. The SAM2 model utilizes machine learning to track an object through time. The SAM2 model outputs a "mask" overlaying the computed area of an object that can be subsequently used by OpenCV to perform conventional Canny edge detection,10 producing a drop profile. The advantage of this machine learning model over conventional methods is edge determination under minimally-distinctive lighting conditions, with low background-to-object contrast. ISS data is captured using a single camera with lighting tailored for PTV rather than edge detection, making machine learning an excellent tool for simultaneously obtaining higher-fidelity automated edge measurements in an environment not well suited to other methods. Rough edges obtained from this method will be further refined by fitting to physics-informed capillary surface equations defined by physically-possible interfacial shapes. In addition to deployment and extraction, development of the refined edge detection model and Surface Evolver code (described below) has the potential to perform shape analysis on the steady-state shearing of the RSD. Surface Evolver has the ability to describe the shape of capillary surfaces by providing a simulated system with known parameters. The software operates as an iterative static solver that seeks the lowest surface energy state (i.e., minimum surface area) for a static fluid system. In comparisons to images from deployment, Surface Evolver is found to very closely resemble the droplet shape during axisymmetric deployment. Modifications to the model to enable analysis of non-axisymmetric cases are currently underway, in order to evaluate an RSD deployment phenomenon where symmetry is lost, and the droplet pins on one side of the static ring before the other.
Bibliography Type:	Description: (Last Updated: 04/15/2026)
Articles in Peer-reviewed Journals	Hirsa AH. "Non-Newtonian flow: Interfacial and bulk phenomena." Fluids. 2026 Mar 28;11(4):91. https://doi.org/10.3390/fluids11040091 , Mar-2026